

29 · ALGEBRA MINI-MODULE

29 · ALGEBRA — ~8–12 Qs in the actual paper (equations, identities, AP/GP, coordinate geometry)

Evidence: 400-Q census of the four Feb 2026 shifts — algebra (equations, identities, AP/GP, coordinate geometry) appears $\approx$ 8–12 times per shift, as single-step templates RRB recycles with fresh numbers. Companion drill: file 22 Q91–110.

UNDERSTAND

- **What appears (~8–12 Qs/shift):** 2–3 word problems (ages, numbers, two-digit numbers — one or two variables), 2–3 quadratics (find k , sum/product of roots, build from roots), 2–3 identities ($a^2 - b^2$, $(a \pm b)^2$, $x + 1/x$), 1–2 AP/GP (n th term, sum, AM/GM), 1–2 coordinate geometry (distance, midpoint, section, slope, area).
- **Level: moderate — recognition, not depth.** Every question in this module dies in ≤ 3 lines. If your solution is longer, you are on the textbook path and paying seconds the paper does not give.
- **Time & scoring:** 60–75 s per question in the exam (60 s in this drill). +1 right, $-\frac{1}{3}$ wrong — a computed answer is worth four coin flips, so recognise-then-compute; never blind-guess.
- **Everything collapses to five engines:** (1) words $\rightarrow$ one equation; (2) sum of roots = $-b/a$, product = c/a ; (3) the three identity patterns; (4) $a_n = a + (n - 1)d$ and $S_n = n/2 \times (\text{first} + \text{last})$; (5) the coordinate toolbox.
- **RRB recycles skeletons:** the father-son age pair, the square-difference ($107^2 - 93^2$ family), the sum-of-AP, the section-ratio point — learn the skeleton, not the digits.
- **Explain-it test:** finish this block able to explain why sum of roots = $-b/a$ (expand $(x - p)(x - q)$) and why $a^2 - b^2 = (a + b)(a - b)$ (the cross terms cancel). If you can, you own this topic.

THE TRICK (fastest exam-hall method)

TRICK A — linear equations (one & two variable; age/number words)

1. Name the asked unknown x ; each sentence becomes one equation ("in t years" $\Rightarrow +t$ to BOTH ages; "a is n times b" $\Rightarrow a = nb$).
2. Sum S & difference D of two numbers: larger = $(S + D) \div 2$, smaller = $(S - D) \div 2$ — one step, no variable.
3. Two-digit number: write $10a + b$; reversed $\Rightarrow 10b + a$. Subtracting gives $(10a + b) - (10b + a) = 9(a - b)$ — the digit difference appears instantly.

Why it works: naming the unknown and translating sentence-by-sentence IS the solution — every age/number problem is one linear equation in disguise.

TRICK B — quadratics (roots, sum/product, build-from-roots)

1. $ax^2 + bx + c = 0 \Rightarrow$ sum of roots = $-b/a$, product = c/a . Fix the signs FIRST — the $-\frac{1}{3}$ lives here.
2. One root r given $\Rightarrow$ other root = (sum) $- r$; then k (or c) = $r \times$ other root.
3. Build from roots p, q : $x^2 - (p + q)x + pq = 0$ — the sum goes with MINUS, the product is the constant.
4. Difference of roots: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$.

Why it works: $(x - p)(x - q)$ expands to $x^2 - (p + q)x + pq$ — the coefficients literally are the sum and the product.

TRICK C — identities ($a^2 - b^2$, $(a \pm b)^2$, $a^3 \pm b^3$, $x + 1/x$)

1. $a^2 - b^2 = (a + b)(a - b)$ — for numeric squares never square first: $107^2 - 93^2 = 200 \times 14 = 2800$.
2. $(a \pm b)^2 = a^2 \pm 2ab + b^2$; $a^2 + b^2 = (a + b)^2 - 2ab = (a - b)^2 + 2ab$.
3. $x + 1/x = k \Rightarrow x^2 + 1/x^2 = k^2 - 2$ and $x^3 + 1/x^3 = k^3 - 3k$. If the stem says $x - 1/x = k$, then $x^2 + 1/x^2 = k^2 + 2$ — the sign flips.
4. $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ — the middle sign is OPPOSITE.

Why it works: squaring $(x + 1/x)$ leaves $x^2 + 1/x^2$ plus exactly 2 — subtract it back.

TRICK D — AP/GP (n th term, sum, AM/GM)

1. AP: $a_n = a + (n - 1)d$; $S_n = n/2 \times (\text{first} + \text{last})$. Term count: $n = (\text{last} - \text{first})/d + 1$.
2. GP: $a_n = a \cdot r^{n-1}$; for short GPs just write the terms and add ($2 + 6 + 18 + 54 = 80$) — usually faster than the formula.
3. AM = $(a + b) \div 2$, GM = $\sqrt{ab}$ — for 4 and 16: AM 10, GM 8. AM $\geq$ GM always.

Why it works: an AP adds a constant each step, a GP multiplies by one — each formula is that pattern written n times.

TRICK E — coordinate geometry (the toolbox)

1. Distance = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ — spot a Pythagorean triple (3-4-5, 6-8-10, 5-12-13, 8-15-17) and skip the root

work.

2. Midpoint = average the coordinates. Section $m_1 : m_2$ from A to B $\Rightarrow x = (m_1x_2 + m_2x_1) \div (m_1 + m_2)$, same for y — m_1 multiplies B (the second point).

3. Slope = $(y_2 - y_1)/(x_2 - x_1)$; line through a point: $y - y_1 = m(x - x_1)$.

4. Area of triangle = $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$; if a side lies on an axis, just $\frac{1}{2} \times \text{base} \times \text{height}$.

Why it works: distance is Pythagoras; midpoint and section are weighted averages — nothing new to memorise.

TRICK F — variation (occasional)

1. Direct $x \propto y \Rightarrow x = ky$ (x/y fixed); inverse $x \propto 1/y \Rightarrow xy = k$.

2. Get k from the given pair, then substitute the asked value. "x doubles when y halves" $\Rightarrow$ inverse.

Why it works: "proportional" means the ratio (or product) is a constant k — find it once, substitute everywhere.

WORKED TEMPLATES (exam-real questions, fully solved)

TEMPLATE 1 — linear, age word (two variables)

A father is four times as old as his son. After 10 years, he will be twice as old as his son. The son's present age is:

A. 5 B. 10 C. 15 D. 20

Answer: A. Son = $x \Rightarrow$ father = $4x$; $4x + 10 = 2(x + 10) \Rightarrow 2x = 10 \Rightarrow x = 5$. Plants: 15 (son in 10 years) and 20 (father now).

TEMPLATE 2 — linear, two-digit number

The sum of the digits of a two-digit number is 9. If 27 is subtracted from the number, its digits get reversed. The number is:

A. 36 B. 45 C. 54 D. 63

Answer: D. $10a + b - 27 = 10b + a \Rightarrow 9(a - b) = 27 \Rightarrow a - b = 3$; with $a + b = 9 \Rightarrow a = 6, b = 3 \Rightarrow 63$. Check: $63 - 27 = 36$, the reverse. Plant A (36) is the reversed number itself.

TEMPLATE 3 — quadratic, find k from one root

If one root of $x^2 - 7x + k = 0$ is 2, then $k =$

A. 10 B. 5 C. 14 D. -10

Answer: A. Plug in: $4 - 14 + k = 0 \Rightarrow k = 10$. Faster: other root = $7 - 2 = 5 \Rightarrow k = 2 \times 5 = 10$. Plant B (5) is the other root.

TEMPLATE 4 — quadratic, sum/product manipulation

If α and β are the roots of $x^2 - 5x + 6 = 0$, then $1/\alpha + 1/\beta =$

A. $6/5$ B. $5/6$ C. $1/6$ D. 5

Answer: B. $1/\alpha + 1/\beta = (\alpha + \beta)/(\alpha\beta) = 5/6$ — sum over product, one step. Plants: 5 (the sum), $1/6$ (one over the product), $6/5$ (flipped).

TEMPLATE 5 — identity, $a^2 - b^2$

The value of $107^2 - 93^2$ is:

A. 1400 B. 2800 C. 3200 D. 280

Answer: B. $(107 + 93)(107 - 93) = 200 \times 14 = 2800$. Plant A (1400) halves the sum; D (280) drops a zero.

TEMPLATE 6 — identity, $x + 1/x$ family

If $x + 1/x = 5$, then $x^2 + 1/x^2 =$

A. 25 B. 21 C. 23 D. 27

Answer: C. $(x + 1/x)^2 = x^2 + 2 + 1/x^2 \Rightarrow x^2 + 1/x^2 = 25 - 2 = 23$. Plant A (25) forgot the -2; D (27) added it instead.

TEMPLATE 7 — AP, sum of n terms

The sum of the first 20 terms of the AP 3, 7, 11, ... is:

A. 780 B. 800 C. 820 D. 840

Answer: C. $a_{20} = 3 + 19 \times 4 = 79$; $S = 20/2 \times (3 + 79) = 10 \times 82 = 820$. Plant D (840) is a term-count slip; A (780) uses $a_{20} = 75$.

TEMPLATE 8 — coordinate geometry, section formula

The point that divides the line segment joining A(1, 1) and B(6, 11) internally in the ratio 2 : 3 is:

A. (5, 3) B. (3, 4) C. (4, 7) D. (3, 5)

Answer: D. $x = (2 \times 6 + 3 \times 1) \div 5 = 3, y = (2 \times 11 + 3 \times 1) \div 5 = 5 \Rightarrow (3, 5)$. The ratio 2 : 3 makes 5 parts; A swaps the coordinates, C swaps the weights (3 and 2).

TRAPS (what costs $-\frac{1}{2}$ here)

- **" +t years" applies to BOTH ages.** Writing $4x + 10 = 2x + 10$ (forgetting the son's own +10) collapses to $x = 0$ — you then guess between the planted 15 (son later) and 20 (father now). Put both "+t" in before

simplifying.

- **Sum of roots is $-b/a$ — signs first.** For $x^2 - 7x + 10$ the sum is $+7$ (the formula's minus cancels the equation's). Product c/a can be negative — opposite-sign roots multiply to a negative.
- **Build-from-roots: the middle term is MINUS the sum.** Roots -2 and $7 \Rightarrow x^2 - 5x - 14 = 0$. The planted pair: $x^2 + 5x - 14$ (sum sign flipped) and $x^2 - 5x + 14$ (product sign forgotten).
- **$a^2 - b^2$: square LAST, never first.** $11449 - 8649$ invites slips; $200 \times 14 = 2800$ takes 5 seconds. Halving the sum ($100 \times 14 = 1400$) is the planted wrong value.
- **$x + 1/x$ vs $x - 1/x$.** $k^2 - 2$ when the stem adds, $k^2 + 2$ when it subtracts — forgetting the ± 2 gives the planted k^2 .
- **AP term count.** $n = (\text{last} - \text{first})/d + 1$ — 5 to 32 with $d = 3$ is 10 terms, not 9 . $S_n = n/2 \times (\text{first} + \text{last})$; the plants include a_n itself and off-by-one- d sums.
- **Coordinate slips.** Distance: the squared sum is planted beside the distance — take the root ($\sqrt{25} = 5$, not 25). Slope: keep the same point first in both differences or the sign flips. Section: divide by $(m_1 + m_2)$ — never $m_1 m_2$ — and m_1 multiplies the far point.
- **GP is not AP.** $3, 6, 12$ grows $\times 2$, not $+3$; "sum of the first 4" of $2, 6, 18$ is $2 + 6 + 18 + 54 = 80$ — the planted 162 is the 5th term, 26 is only the first three terms.

SPEED DRILL (10 Qs, 60 s/Q)

Q1. The sum of two numbers is 36 and one of them is twice the other. The smaller number is:

A. 12 B. 24 C. 18 D. 9

Q2. A mother is three times as old as her daughter. After 8 years, she will be twice as old as her daughter. The daughter's present age (in years) is:

A. 10 B. 8 C. 16 D. 12

Q3. The product of the roots of $3x^2 - 11x + 6 = 0$ is:

A. $11/3$ B. 3 C. 2 D. 6

Q4. The quadratic equation whose roots are -2 and 7 is:

A. $x^2 + 5x - 14 = 0$ B. $x^2 - 5x + 14 = 0$ C. $x^2 + 5x + 14 = 0$ D. $x^2 - 5x - 14 = 0$

Q5. The value of $104^2 - 96^2$ is:

A. 800 B. 1600 C. 3200 D. 160

Q6. If $x + 1/x = 6$, then $x^2 + 1/x^2 =$

A. 36 B. 32 C. 34 D. 38

Q7. The 10th term of the AP $5, 8, 11, \dots$ is:

A. 32 B. 35 C. 29 D. 38

Q8. The sum of the first four terms of the GP $2, 6, 18, \dots$ is:

A. 26 B. 54 C. 162 D. 80

Q9. The distance between the points $(2, 3)$ and $(5, 7)$ is:

A. 5 B. 4 C. 7 D. 25

Q10. The midpoint of the line segment joining $(-3, 4)$ and $(5, -2)$ is:

A. $(-4, 1)$ B. $(1, 1)$ C. $(2, 2)$ D. $(1, 3)$

Answer key

1. **A** — $x + 2x = 36 \Rightarrow 3x = 36 \Rightarrow x = 12$ (smaller).
2. **B** — $3x + 8 = 2(x + 8) \Rightarrow x = 8$.
3. **C** — product $= c/a = 6/3 = 2$.
4. **D** — sum 5 , product $-14 \Rightarrow x^2 - 5x - 14 = 0$.
5. **B** — $(104 - 96)(104 + 96) = 8 \times 200 = 1600$.
6. **C** — $k^2 - 2 = 36 - 2 = 34$.
7. **A** — $a_{10} = 5 + 9 \times 3 = 32$.
8. **D** — $r = 3: 2 + 6 + 18 + 54 = 80$.
9. **A** — $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.
10. **B** — $((-3 + 5) \div 2, (4 - 2) \div 2) = (1, 1)$.